Ex. 8.1 Q18

$$\int \frac{2^{\sqrt{y}}}{2\sqrt{y}}\,dy$$

(A function of the form $\int \frac{f(\sqrt{y})}{\sqrt{y}}\,dy$)

$u = \sqrt{y}$

$du = \frac{1}{2\sqrt{y}}\,dy$

Try $u = \sqrt{y}$

$du = \frac{1}{2\sqrt{y}}\,dy$

$$\text{Integral} = \int 2^u\,du = \frac{2^u}{\ln 2} + C$$

$$= \frac{2^{\sqrt{y}}}{\ln 2} + C$$

Q6 $\displaystyle\int \frac{dx}{x-\sqrt{x}}$

Attempt $u = x - \sqrt{x}$

$du = 1 - \frac{1}{2\sqrt{x}}\, dx$

$$= \int \frac{dx}{\sqrt{x}(\sqrt{x}-1)}$$

$\left(\frac{f(\sqrt{x})}{\sqrt{x}} \quad f(t) = \frac{1}{t-1} \right)$

$$= \int \frac{2\,du}{u-1}$$

$v = \sqrt{x}$

$du = \frac{1}{2\sqrt{x}}\, dx$

$$= 2\int \frac{1}{u-1}\, du$$

$$= 2\ln|u-1| + c$$

$$= 2\ln|\sqrt{x}-1| + c$$

Q 26

$$\int \frac{6}{\sqrt{y}(1+y)}\,dy \qquad \left(\frac{f(\sqrt{y})}{\sqrt{y}}\right)$$

$$f(t) = \frac{6}{1+t^2}$$

write it as $1+(\sqrt{y})^2$

$u = \sqrt{y} \Rightarrow y = u^2$

$du = \frac{1}{2\sqrt{y}}\,dy$

$$\text{Integral} = \int \frac{6\cdot 2}{1+u^2}\,du$$

$$= 12\int \frac{1}{1+u^2}\,du$$

$$= 12 \arctan u + C$$

$$= 12 \arctan \sqrt{y} + C$$

Idea:
If we are given $\pm 1 \pm f(x)$ in the integrand, purposely write it as $\pm 1 \pm (\sqrt{f(x)})^2$

Q40 $\displaystyle\int \frac{\sqrt{x}}{1+x^3}\,dx$ Hint : $u = x^{3/2}$.

$1+x^3 = 1+(x^{\frac{3}{2}})^2$

$= 1+(x^{\frac{3}{2}})^2$

$u = x^{\frac{3}{2}}$

$du = \frac{3}{2}\sqrt{x}\,dx$

$$\text{Integral} = \frac{2}{3}\int \frac{1}{1+u^2}\,du$$

$$= \frac{2}{3}\arctan u + C$$

$$= \frac{2}{3}\arctan x^{\frac{3}{2}} + C$$

Q25 $\displaystyle\int \frac{dy}{\sqrt{e^{2y}-1}}$

$e^{2y}-1 = (e^y)^2 - 1$

$u = e^y$

$du = e^y\,dy$

$= u\,dy$

$$= \int \frac{1}{u\sqrt{u^2-1}}\,du$$

$$= \operatorname{arcsec}|u| + C$$

$$= \operatorname{arcsec} e^y + C$$

#Q27 (similar)

Q19 $\displaystyle\int_1^2 \frac{8}{x^2 - 2x + 2}\,dx$ $\rightarrow \pm 1 \pm (f(x))^2$

$$= \int_1^2 \frac{8}{\underbrace{(x-1)^2+1}_{u^2+1}}\,dx$$

Q38 $\displaystyle\int \frac{d\theta}{\cos\theta - 1}$ $\qquad \cos\theta - 1 = (\sqrt{\cos\theta})^2 - 1$

$\sin^2\theta + \cos^2\theta = 1$

$A^2 - B^2 = (A-B)(A+B)$

$\cos^2\theta - 1 = -\sin^2\theta$

-1

$$\int \frac{d\theta}{\cos\theta - 1}$$

$$= \int \frac{d\theta}{\cos\theta - 1} \cdot \frac{\cos\theta + 1}{\cos\theta + 1}$$

$$= \int \frac{\cos\theta + 1}{-\sin^2\theta}\,d\theta$$

$$= -\int \underbrace{\frac{\cos\theta}{\sin^2\theta}}_{\underbrace{\frac{\cos\theta}{\sin\theta}\cdot\frac{1}{\sin\theta}}_{\cot\theta\csc\theta}}\,d\theta - \int \underbrace{\frac{1}{\sin^2\theta}}_{\csc^2\theta}\,d\theta$$

$$= \csc\theta + \cot\theta + C$$

$\underline{sst}$ $\underline{tss}$

$$\frac{d}{dx}(\sec x) = \sec x \tan x$$

$$\int \sec x \, dx = \ln|\underline{\sec x} + \underline{\tan x}| + C$$

$$\int \sec^2 x \, dx = \tan x + C$$

$$\int \tan x \sec x \, dx = \sec x + C$$

$$\int \underline{\tan x} \, dx = \ln|\underline{\sec x}| + C$$

$$\ln|\sec x + \sec x|$$

$$\ln|\underline{\underline{2}} \sec x|$$

$$\int \cot x \, dx = -\ln|\csc x| + C$$

$\cancel{\int \csc}$

Q44

$$\int \frac{\tan\theta + 3}{\sin\theta}\, d\theta$$

$$= \int \underbrace{\frac{\cancel{\sin\theta}}{\cos\theta}}_{\sec\theta} \cdot \underbrace{\frac{1}{\cancel{\sin\theta}}}_{\cancel{\csc}}\, d\theta + \int \underbrace{\frac{3}{\sin\theta}}_{\csc\theta}\, d\theta$$

$$= \ln|\sec\theta + \tan\theta| - 3\ln|\csc\theta + \cot\theta|$$

$$+ C$$